

Question - 1 (5 marks)

Write formal proof of the following theorem.

Theorem: If A is an $n \times n$ matrix then the following statements are equivalent.

- (i) A is non-singular.
- (ii) The *only* possible solution to $A\mathbf{x} = \mathbf{0}$ is the trivial solution.
- (iii) A is row equivalent to I_n .
- (iv) $A\mathbf{x} = \mathbf{b}$ is consistent with a unique solution $\forall \mathbf{b} \in \mathbb{R}^n$.
- (v) $\det(A) \neq 0$.

Proof:

Statement i implies statement ii

If A is nonsingular and $\bar{\mathbf{x}}$ is solution of $A\bar{\mathbf{x}} = \bar{\mathbf{0}}$

Then

$$\bar{\mathbf{x}} = I \bar{\mathbf{x}} = (A^{-1}A)\bar{\mathbf{x}} = A^{-1}(A\bar{\mathbf{x}}) = A^{-1}(\bar{\mathbf{0}}) = \mathbf{0}$$

Thus $A\bar{\mathbf{x}} = \bar{\mathbf{0}}$ has only the trivial solution. *proved*

Statement ii implies statement iii

If we apply elementary row operations to reduce the system $A\bar{\mathbf{x}} = \bar{\mathbf{0}}$ into an equivalent system $U\bar{\mathbf{x}} = \bar{\mathbf{0}}$ ($E_1 E_2 \dots E_k A = U$) where U is in row echelon form. If one of the diagonal element of U were zero, then last row of U would be consist entirely of 0's. However, then $A\bar{\mathbf{x}} = \bar{\mathbf{0}}$ would be equivalent to the system with more unknown than equations, and hence, an $m \times n$ homogeneous system of linear equations has a nontrivial solution if $n > m$. Thus, U must be a strictly triangular matrix with diagonal elements all equal to 1. It then follow I is the reduced row echelon form of A and hence A is row equivalent to I . *proved*

Statement iii implies iv

If A is row equivalent to I then $E_1 E_2 \dots E_k A = I$

Now let $\bar{\mathbf{x}}$ be any solution of $A\bar{\mathbf{x}} = \bar{\mathbf{b}}$,

Consider the augmented system $[A | \mathbf{b}]$

Then, applying the same sequence of elementary matrix multiplication leads to

$$\begin{aligned} & (E_1 E_2 \dots E_k) [A, \bar{b}] \\ \Rightarrow & [(E_1 E_2 \dots E_k) A, (E_1 E_2 \dots E_k) \bar{b}] \\ \Rightarrow & [I, \bar{c}] \end{aligned}$$

$\Rightarrow \bar{x} = \bar{c}$ is the only possibility.

$\Rightarrow \bar{x}$ is a unique solution $\forall b \in \mathbb{R}^n$

proved

Statement- iv implies v

If $A\bar{x} = \bar{b}$ is consistent with a unique solution $\forall \bar{b} \in \mathbb{R}^n$

and A is row equivalent to I

i.e. $E_k E_{k-1} \dots E_2 E_1 A = I$

or $A = E_1^{-1} E_2^{-1} \dots E_{k-1}^{-1} E_k^{-1} I$

$$\det(A) = \det(E_1^{-1} E_2^{-1} \dots E_{k-1}^{-1} E_k^{-1} I)$$

$$= \det(E_1^{-1}) \det(E_2^{-1}) \dots \det(E_k^{-1}) \det(I)$$

$\because E_i$'s are elementary matrices and all E_i 's & E_i^{-1} 's are nonsingular

$$\therefore \det(E_i^{-1}) \neq 0 \quad \text{and} \quad \det(I) \neq 0$$

Hence $\det(A) \neq 0$

proved

Statement- v implies i

If $\det(A) \neq 0$ and let U is the row echelon form of A

Then

$$U = E_k E_{k-1} \dots E_2 E_1 A$$

and $\det(U) = \det(E_k E_{k-1} \dots E_2 E_1 A)$

$$\Rightarrow \det(U) = \det(E_k) \det(E_{k-1}) \dots \det(E_2) \det(E_1) \det(A)$$

Since $\det(E_i^{-1}) \neq 0$ as E_i 's are elementary matrices

and $\det(A) \neq 0$ (given)

Then $\det(U) \neq 0$ and hence U is in strictly triangular form and all the diagonal entries are nonzero and the reduced row echelon form of A must be an identity matrix. Hence A is invertible.

proved

Question - 2 (5 marks)

Write formal proof of the following theorem, by asserting formal results regarding determinants from the theory of *Elementary Matrices*.

Theorem: Let A be a square matrix of order $n \times n$.

- (i) If B is the matrix that results from multiplying a row of A by a scalar, c , then $\det(B) = c \det(A)$.
- (ii) If B is the matrix that results from interchanging two rows of A , then $\det(B) = -\det(A)$.
- (iii) If B is the matrix that results from adding a multiple of one row of A to another row of A , then $\det(B) = \det(A)$.

i/ If B is the matrix that results from multiplying a row of A by a scalar, c ,

i.e $B = (\text{Elementary matrix of type I}) A$ (say E_1)
 $B = E_1 A$

As E_1 is an identity matrix whose one row is multiplied by a scalar c .

Then $\det(E_1) = c \det(I) = c$

Now $\det(B) = \det(E_1 A)$
 $= \det(E_1) \det(A)$
 $= c \det(A)$ *proved*

ii/ If B is the matrix results from interchanging two rows of A .

i.e $B = (\text{Elementary matrix of type II}) A$ (say E_2)
 $B = E_2 A$

As E_2 is an identity matrix whose two rows are interchanged

Then $\det(E_2) = -1 \det(I) = -1$

Now, $\det(B) = \det(E_2 A)$
 $= \det(E_2) \det(A)$
 $= -\det(A)$ *proved*

iii/ If B is the matrix that results from adding a multiple of one row of A to another row of A

i.e $B = (\text{Elementary matrix of type III}) A$ (say E_3)
 $B = E_3 A$

As E_3 is an identity matrix whose one row is the sum of its own and a nonzero multiple of another row.

Hence a triangular matrix with all diagonal elements are 1, therefore the determinant is also 1.

$$\text{Hence } \det(E_3) = 1 \det(I) = 1 \quad (\text{i.e. unchanged})$$

$$\begin{aligned} \text{Now, } \det(B) &= \det(E_3 A) \\ &= \det(E_3) \det(A) = \det(A) \quad \text{proved} \end{aligned}$$

Question - 3 (5 marks)

Let $\mathcal{R}$ denote the set of real numbers. Define scalar multiplication by:

$$\alpha x = \alpha \cdot x \quad (\text{the usual multiplication of real numbers})$$

and define addition, denote $\oplus$, by

$$x \oplus y = \max(x, y) \quad (\text{the maximum of the two numbers})$$

Is $\mathcal{R}$ a vector space with these operations? Formally prove your answer.

Let $\mathcal{R}$ denotes the set of real numbers.

Let x, y , and $z \in \mathcal{R}$ and α , and β be any scalars $\in \mathbb{R}$

Axiom 1: $x + y = y + x$ for any x and y in $\mathcal{R}$

$$x \oplus y = \max(x, y) = \begin{cases} x & \text{when } x > y \\ y & \text{otherwise} \end{cases}$$

$$y \oplus x = \max(y, x) = \begin{cases} x & \text{when } x > y \\ y & \text{otherwise} \end{cases}$$

$$\text{Hence } x \oplus y = y \oplus x$$

$\therefore$ Axiom 1 holds.

Axiom 2: $(x + y) + z = x + (y + z)$ for any x, y , and z in $\mathcal{R}$

$$\begin{aligned} (x \oplus y) \oplus z &= \max(x, y) \oplus z = \left(\begin{cases} x & x > y \\ y & \text{otherwise} \end{cases} \right) \oplus z \\ &= \begin{cases} x & x > y > z \\ y & y > x > z \\ z & z > x > y \end{cases} \end{aligned}$$

$$\text{Similarly, } x \oplus (y \oplus z) = \begin{cases} x & x > y > z \\ y & y > x > z \\ z & z > x > y \end{cases}$$

$$\text{Hence } (x \oplus y) \oplus z = x \oplus (y \oplus z)$$

$\therefore$ Axiom 2 holds.

Axiom 3: There exist an element 0 in $\mathbb{R}$ such that $x + 0 = x = 0 + x$ for each x in $\mathbb{R}$

$$x \oplus 0 = 0 \oplus x = \max(x, 0)$$

$$\neq x \quad \text{if } x \text{ is -ve}$$

$\therefore$ Axiom 3 fails to hold.

Axiom 4: For each $x \in \mathbb{R}$ there exist an element $-x$ in $\mathbb{R}$

$$\text{such that } x + (-x) = 0$$

$$x \oplus (-x) = \max(x, -x) = (-x) \oplus x$$

$$\neq 0$$

Axiom 4 fails to hold.

Axiom 5: $\alpha(x+y) = \alpha x + \alpha y$ for each scalar α and x and $y \in \mathbb{R}$

$$\alpha(x \oplus y) = \alpha[\max(x, y)]$$

$$= \alpha(x)$$

$$= \alpha x$$

$$\alpha x \oplus \alpha y = \max(\alpha x, \alpha y)$$

$$\neq \alpha x \quad \text{when } \alpha \text{ is -ve}$$

$\therefore$ Axiom 5 fails to hold.

Axiom 6: $(\alpha + \beta)x = \alpha x + \beta x$ for any scalar α and β and $x \in \mathbb{R}$

$$(\alpha + \beta) \cdot x = (\alpha + \beta)x$$

$$\text{and } \alpha \cdot x + \beta \cdot x = \alpha x + \beta x$$

$$= \max(\alpha x, \beta x) = \alpha x \quad \text{if } \alpha > \beta$$

$$\neq (\alpha + \beta)x$$

$\therefore$ Axiom 6 fails to hold

Axiom 7: $(\alpha\beta)x = \alpha(\beta x)$ for any scalar α and β and $x \in \mathbb{R}$

$$(\alpha\beta) \cdot x = (\alpha\beta)x = \alpha\beta x$$

$$\text{and } \alpha \cdot (\beta \cdot x) = \alpha \cdot (\beta x) = \alpha\beta x$$

$$\text{Hence } (\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x)$$

$\therefore$ Axiom 7 holds.

Axiom 8 : $1 \cdot x = x$ for $x \in \mathbb{R}$

$$1 \cdot x = 1x = x$$

$\therefore$ Axiom 8 holds.

Axiom 3, 4, 5, and 6 all fails to hold.

Hence $\mathbb{R}$ is not a vector space with the given operations.

Question - 4 (5 marks)

Let

$$A = \begin{bmatrix} x & 1 & 1 \\ 1 & x & -1 \\ -1 & -1 & x \end{bmatrix}$$

(a) Compute the value of $\det(A)$. (Your answer should be a function of x).

(b) For what values of x will the matrix be singular? Explain concisely.

a/

$$A = \begin{bmatrix} x & 1 & 1 \\ 1 & x & -1 \\ -1 & -1 & x \end{bmatrix}$$

$$\det(A) = x \begin{vmatrix} x & -1 \\ -1 & x \end{vmatrix} - 1 \begin{vmatrix} 1 & -1 \\ -1 & x \end{vmatrix} + 1 \begin{vmatrix} 1 & x \\ -1 & -1 \end{vmatrix}$$

$$= x(x^2 - 1) - 1(x - 1) + 1(-1 + x)$$

$$= x(x^2 - 1) - 1(x - 1) + 1(x - 1)$$

$$\det(A) = x(x^2 - 1)$$

Ans

b/ If $\det(A) = 0$ then A will be singular

i.e. $x(x^2 - 1) = 0$

$$\Rightarrow \text{either } x = 0 \text{ or } x^2 - 1 = 0$$

$$\Rightarrow x = \pm 1$$

Therefore For $x = 0$ or $x = 1$ or $x = -1$

The given matrix A will be singular

Ans

Question - 5 (10 marks)

Matrix M in partitioned form, and decomposed as $M = UDL$ via Schur Complement is given as:

$$M = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{cc} I_p & BD^{-1} \\ 0 & I_q \end{array} \right] \left[\begin{array}{cc} A - BD^{-1}C & 0 \\ 0 & D \end{array} \right] \left[\begin{array}{cc} I_p & 0 \\ D^{-1}C & I_q \end{array} \right]$$

where

$$A = \begin{bmatrix} 5 & 9 \\ -5 & -3 \end{bmatrix}; \quad B = \begin{bmatrix} 2 \\ -1 \end{bmatrix}; \quad C = \begin{bmatrix} 3 & 4 \end{bmatrix}; \quad D = [1]; \quad I_p = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad I_q = [1];$$

(a) Calculate a partitioned matrix expression for M^{-1} using the given Schur Complement based decomposition.

(b) Using this expression, determine M^{-1} .

9/ As $M = UDL$

$$M = \left[\begin{array}{cc} I_p & BD^{-1} \\ 0 & I_q \end{array} \right] \left[\begin{array}{cc} A - BD^{-1}C & 0 \\ 0 & D \end{array} \right] \left[\begin{array}{cc} I_p & 0 \\ D^{-1}C & I_q \end{array} \right]$$

Taking inverse both sides

$$M^{-1} = (UDL)^{-1} = L^{-1} D^{-1} U^{-1}$$

$$= \left[\begin{array}{cc} I_p & 0 \\ D^{-1}C & I_q \end{array} \right]^{-1} \left[\begin{array}{cc} A - BD^{-1}C & 0 \\ 0 & D \end{array} \right]^{-1} \left[\begin{array}{cc} I_p & BD^{-1} \\ 0 & I_q \end{array} \right]^{-1}$$

$$M^{-1} = \left[\begin{array}{cc} I_p & 0 \\ -D^{-1}C & I_q \end{array} \right] \left[\begin{array}{cc} (A - BD^{-1}C)^{-1} & 0 \\ 0 & D^{-1} \end{array} \right] \left[\begin{array}{cc} I_p & -BD^{-1} \\ 0 & I_q \end{array} \right]$$

Ans

b/ As $D = [1] \Rightarrow D^{-1} = [1]$

$$D^{-1}C = [1] \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

and

$$BD^{-1} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} [1] = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Now

$$BD^{-1}C = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ -3 & -4 \end{bmatrix}$$

$$A - BD^{-1}C = \begin{bmatrix} 5 & 9 \\ -5 & -3 \end{bmatrix} - \begin{bmatrix} 6 & 8 \\ -3 & -4 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$$

$$(A - BD^T c)^{-1} = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}^{-1} \quad \left| \quad \begin{array}{l} \text{So, } \det(A - BD^T c) = -1 + 2 = 1 \\ \text{adj}(A - BD^T c) = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \end{array} \right.$$

$$(A - BD^T c)^{-1} = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$$

$$M^{-1} = \begin{bmatrix} I_p & 0 \\ -D^T c & I_q \end{bmatrix} \begin{bmatrix} (A - BD^T c)^{-1} & 0 \\ 0 & D^{-1} \end{bmatrix} \begin{bmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & -4 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 0 \\ -11 & 7 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M^{-1} = \left[\begin{array}{cc|c} 1 & -1 & -3 \\ 2 & -1 & -5 \\ \hline -11 & 7 & 30 \end{array} \right] \quad \text{Ans}$$

Question - 6 (5 marks)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

- (a) Compute the determinant of A . Is A singular?
 (b) Compute $\text{adj}(A)$ and the product $A \text{adj}(A)$.

$$\text{ay} \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \approx \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix} \approx \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \det(A) = 1 \times 1 \times 0 = 0$$

$\therefore \det(A) = 0$
and hence A is singular. Ans

$$\rightarrow \text{Adj}(A) = [A_{ij}]^t$$

where

$$A_{ij} = (-1)^{i+j} |M_{ij}|$$

$$A_{11} = (-1)^{1+1} |M_{11}| = \begin{vmatrix} 3 & 4 \\ 4 & 5 \end{vmatrix} = 15 - 16 = -1$$

$$A_{12} = (-1)^{1+2} |M_{12}| = -1 \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = -(10 - 12) = 2$$

$$A_{13} = (-1)^{1+3} |M_{13}| = \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = 8 - 9 = -1$$

$$A_{21} = (-1)^{2+1} |M_{21}| = -1 \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -(10 - 12) = 2$$

$$A_{22} = (-1)^{2+2} |M_{22}| = \begin{vmatrix} 1 & 3 \\ 3 & 5 \end{vmatrix} = 5 - 9 = -4$$

$$A_{23} = (-1)^{2+3} |M_{23}| = -1 \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -(4 - 6) = 2$$

$$A_{31} = (-1)^{3+1} |M_{31}| = \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = 8 - 9 = -1$$

$$A_{32} = (-1)^{3+2} |M_{32}| = -1 \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = -(4 - 6) = 2$$

$$A_{33} = (-1)^{3+3} |M_{33}| = \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 3 - 4 = -1$$

$\therefore$

$$\text{Adj}(A) = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

Now

$$A \operatorname{adj}(A) = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

$$A \operatorname{adj} A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Ans